INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

GLOVES, MEN'S AND WOMEN'S, INTERMEDIATE COLD/WET

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the men's and women's intermediate cold/wet gloves constructed with stretch fabric on the back of the hand, leather palm, waterproof insert and an insulated lining.

1.2 Classification. This specification covers the following glove sizes.

1.2.1 Sizes.

Small
Medium
Medium – Wide
Large
Large – Wide
X-Large
X-Large – Wide

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AMSC N/A FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199 - Thread, Polyester Core Cotton- or Polyester-Covered
A-A-59826 - Thread, Nylon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-5664 - Webbing, Textile, Elastic
MIL-PRF-17337 - Webbing, Textile, Woven Nylon
MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

BUREAU OF ALCOHOL, TOBACCO AND FIREARMS, DEPARTMENT OF THE TREASURY

Formulas for Denatured Alcohol (27 CFR Part 21)

(Copies of referenced document area available online at https://www.atf.gov.)

ENVIRONMENTAL PROTECTION AGENCY

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) - (40 CFR Part 162)
State Registration of Pesticide Products

(Copies of this documents are available online at https://www.epa.gov/pesticides.)

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FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP8	-	Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9	-	Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8	-	Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM20	-	Test Method for Fiber Analysis:Qualitative
AATCC TM20A	-	Test Method for Fiber Analysis: Quantitative
AATCC TM22	-	Test Method for Water Repellency: Spray
AATCC TM135	-	Test Method for Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D471	-	Standard Test Method for Rubber Property-Effect of Liquids
ASTM D751	-	Standard Test Method for Coated Fabrics
ASTM D1388	-	Standard Test Method for Stiffness of Fabrics
ASTM D1424	-	Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D1777	-	Standard Test Method for Thickness of Textile Materials
ASTM D1814	-	Standard Test Method for Measuring Thickness of Leather Units
ASTM D1876	-	Standard Test Method for Peel Resistance of Adhesives (T-Peel Test)
ASTM D2209	-	Standard Test Method for Tensile Strength of Leather
ASTM D2261	-	Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure, (Constant-Rate-of-Extension Tensile Testing Machine)
ASTM D2617	-	Standard Test Method for Total Ash in Leather

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ASTM D2807	-	Standard Test Method for Chromic Oxide in Leather (Perchloric Acid Oxidation)
ASTM D2810	-	Standard Test Method for pH of Leather
ASTM D2821	-	Standard Test Method for Measuring the Relative Stiffness of Leather By means of a Torsional Wire Apparatus
ASTM D3495	-	Standard Test Method for Hexane Extraction of Leather
ASTM D3776/D3776M	-	Standard Test Method for Mass Per Unit Area (Weight) of Woven Fabric
ASTM D3787	-	Standard Test Method for Bursting Strength of Textiles-Constant-Rate-of-Traverse(CRT) Ball Burst Test
ASTM D3884	-	Standard Test Method for Abrasion Resistance of Textile Fabrics (Rotary Platform, Double-Head Method)
ASTM D3886	-	Standard Test Method for Abrasion Resistance of Textile Fabrics (Inflated Diaphragm Method)
ASTM D4964	-	Standard Test Method for Tension and Elongation of Elastic Fabrics (Constant-Rate-of-Extension Type Tensile Testing Machine)
ASTM D5035	-	Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)
ASTM D5052	-	Standard Test Method for Permeability of Leather to Water Vapor
ASTM D6076	-	Standard Test Method for Shrinkage Temperature of Leather
ASTM D6193	-	Standard Practice for Stitches and Seams
ASTM E96/E96M	-	Standard Test Method for Water Vapor Transmission of Materials
ASTM F392/F392M	-	Standard Practice for Conditioning Flexible Barrier Materials for Flex Durability
ASTM F903	-	Standard Test Method for Resistance of Materials Used in Protective Clothing to Penetration by Liquids

(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

SAE INTERNATIONAL

SAE-AMS-STD-595, Color Chip 24165 and Color Chip 20150.

(Copies of this document are available online at https://www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Unless a specific exemption has been obtained, nothing in this document, supersedes applicable laws and regulations.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Shade standard. Unless otherwise indicated, the finished glove (materials and components) shall be Foliage Green 504 and shall match color Chip 24165 and for Coyote 498 color Chip 20150 of SAE-AMS-STD-595.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design. The gloves have a stretch fabric on the back of the hand, leather palms, waterproof inserts and an insulated lining. The gloves shall be slip-on, inseamed fourchette design with a two pattern piece thumb and a fleece snow cuff, containing a leather/fabric shell, waterproof/moisture vapor permeable insert, microfiber insulation with brushed polyester/nylon lining, buckle/strap adjustment on the back of the wrist with an elastic take up opposite, and a combination hook and a loop ribbon for attaching gloves to each other.

3.5 Figures. Figures 1-3 are furnished for information purposes only. If there are any inconsistencies between the specification and the figures, the specification shall govern.

3.6 Materials.

3.6.1 Goatskin leather. The leather shall be full grain, chrome tanned, goatskin.

3.6.1.1 Thickness. The thickness of the goatskin leather shall be 2.0 (+ 0.25, -0.50) ounces when tested as specified in 4.4.1.

3.6.1.2 Color. The color of the finished leather shall be Foliage Green 504, Coyote 498 or as specified in the contract and shall be drum dyed, struck through from grain to flesh or as specified in the contract or purchase order (see 6.2) when tested as specified in 4.4.1.

3.6.1.3 Colorfastness to crocking. The leather shall show colorfastness to dry and wet crocking equal to or better than the standard sample. When no standard sample is available the (leather) shall show a minimum rating of 4.0 for dry crocking and minimum rating of 3.0 for wet crocking when tested as specified in 4.4.1.

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3.6.1.4 Finish. The leather shall be soft and pliable. Application of a finish to the grain surface shall be prohibited. If necessary, the flesh side of the leather shall be dry buffed and shaved to obtain a smooth, clean surface and uniform thickness in all areas. Minor variations in color on the flesh side resulting from buffing or shaving are permissible (see 4.4).

3.6.1.5 Water resistant. The leather shall be treated with a water resistant compound to meet the requirements of this document.

3.6.1.6 Stitch tearing strength. At least 80 percent of the specimens tested shall have a stitch tearing strength of not less than 17-pounds for the leather when tested as specified in 4.4.1.

3.6.1.7 Elongation. At least 80 percent of the specimens tested shall have an elongation of not less than 25 percent at a load of 25 pounds. Any specimen which ruptures or exhibits grain crack below 25-pounds shall be reported as a failing specimen. Testing shall be tested as specified in 4.4.1.

3.6.1.8 Shrinkage temperature. The leather shall not shrink before the temperature reaches 98 (± 0.5)°C, nor shall the leather shrink when subjected to a temperature of 98 (± 0.5)°C for a period of 30 seconds when tested as specified in 4.4.1.

3.6.1.9 Stiffness. At least 80 percent of the specimens tested shall have maximum stiffness values of 120 degrees before soaking and 135 degrees after soaking when tested as specified in 4.4.1.

3.6.1.10 Water absorption resistance. At least 80 percent of the specimens tested shall gain not more than 20 percent of their weight by water absorption when tested as specified in 4.4.1.

3.6.1.11 Moisture Vapor Transmission Rate (MVTR). The leather shall meet component requirements for MVTR. At least 80 percent of the specimens tested shall have a MVTR value of not less than 500 grams/square meter/24hours when tested as specified in 4.4.1.

3.6.1.12 Chemical requirements. The leather shall conform to the chemical requirements in Table I when tested as specified in 4.4.1.

TABLE I. Chemical requirements.

Characteristics	Minimum	Maximum
Hexane soluble material, percent 1/	--	25.0
Chromic oxide, percent 1/	3.0	--
pH value	3.3	--
Total ash, percent 1/	--	9.0

1/ Calculated on moisture-free basis.

3.6.2 Back hand fabric.

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3.6.2.1 Fabric, back of hand. The fabric shall be made of a highly water repellent, 4 way stretch material consisting of 91 (± 2) %, nylon and 9 (± 2) % spandex. The finished cloth shall conform to the requirements specified in Table II when tested as specified in 4.4.1. The color for the fabric, back of hand, shall be Foliage Green 504, Coyote 498 or as specified in the contract (see 6.5).

TABLE II. Fabric, back of hand, requirements.

Characteristic	Requirement
Weight, oz./sq. yd.	6.5 (± 0.3)
Stretch, percent Warp Filling	 20-30 20-30
Water Repellency, minimum	100, 100, 90
Colorfastness to Crocking, rating (minimum) Wet Dry	 4.0 4.0
Tear Strength, pounds, (minimum) Warp Filling	 22 9

3.6.3 Nosewipe. The composition of the nosewipe material shall consist of a 100 percent (%) Nylon facing fabric and a base material made of 50 (± 2) % polyester, 50 (± 2) % cotton. The color for the nosewipe material shall be Foliage Green 504, Coyote 498 or as specified in the contract. The nosewipe material shall meet the requirements in Table III when tested as specified in 4.4.1 (see 6.10).

TABLE III. Nosewipe composition material requirements.

Characteristic	Requirement
Weight, oz./sq. yd.	6.6 (± 0.3)
Colorfastness to Crocking, rating, (minimum) Wet Dry	 4.0 5.0
Tear Strength, pounds, (minimum)	3.0

3.6.4 Fleece snow cuff. The fabric for the fleece snow cuff shall be a polyester/nylon/spandex, stretch fleece consisting of a fiber blend of a blend of 53 (±2%) polyester, 38 (± 2) % nylon and 9 (±2) % spandex. The fabric shall be treated with durable water repellency finish. The color shall be Foliage Green 504, Coyote 498 or as specified in

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the contract. The fabric shall meet the requirements in Table IV when tested as specified in 4.4.1. The fleece side of the fabric should face the hand (see 6.6).

TABLE IV. Fleece snow cuff requirements.

Characteristic	Requirement
Weight, oz./sq. yd.	6.4 - 7.8
Colorfastness to Crocking, rating (minimum) Wet Dry	 3.0 4.0
Stretch, percent, (minimum) Width Length	 75 60
Dimensional Stability, percent, (minimum) Length Width	 5.0 5.0
Water repellency, (minimum)	80, 80, 80

3.6.5 Lining.

3.6.5.1 Backing and facing lining fabrics. The backing fabric shall be a 2-bar tricot knitted with nylon monofilament. The facing fabric shall be a fabric consisting of microdenier polyester and Hydrophil® nylon. The fabric shall meet the requirements in Table V when tested as specified in 4.4.1. The polyester facing fabric shall be against the hand and shall be napped in such a way that is comfortable and not clinging to the hand.

3.6.5.2 Index finger lining fabric. The index finger shall be lined with a thin fabric lining conforming to the requirements for the facing fabric in 3.6.5.1 and Table V. The index finger shall have no insulation batting; facing fabric may be laminated to itself for the index finger lining fabric. Testing shall be as specified in 4.4.1.

TABLE V. Finger lining fabrics requirements.

Characteristic	Requirement		
	Backing	Facing	Index Finger
Color	White/Natural	Black/Natural	Black/Natural
Weight, oz./sq. yd.	0.4 - 0.8	2.5 - 3.0	4.0 - 6.5
Bursting Strength, pounds, (minimum)	20	40	40

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3.6.5.3 Finished lining. The finished lining (backing fabric/insulation batting/facing fabric laminate) shall be constructed by bonding the backing fabric to one side of the insulation batting and bonding the un-napped side of the facing fabric to the other side of the insulation batting. The finished lining shall meet the requirements in Table VI when tested as specified in 4.4.1.

TABLE VI. Finished lining requirements.

Characteristic	Minimum	Maximum
Thickness (inch) @ 0.002 psi	0.25	0.36
Drape stiffness, inches: Machine Cross Machine	 -- --	 2.5 2.5
Water vapor transmission, grams/square meter/24 hours	650	--

3.6.6 Insert.

3.6.6.1 Waterproof insert. The waterproof insert shall be constructed from a waterproof, moisture vapor permeable, two or three-layer laminate (see 3.6.6.1.1 thru 3.6.6.1.2). The insert shall be properly sized to fit around the glove insulation. The insert shall be designed so that it can be securely affixed inside the glove, such that it cannot be pulled out when the hand is removed. The finished insert shall be manufactured in such a way that it meets the requirements in Table VII when tested as specified in 4.4.1. Take special note that the insert has no distinction between face and back and may be accidentally inverted in use; refer to end item for proper side to test (see 6.7).

3.6.6.1.1 Two-layer laminate. The two-layer laminate shall consist of a non-woven textile layer and a membrane layer.

3.6.6.1.2 Three-layer laminate. The three-layer laminate shall consist of a non-woven textile layer, a membrane layer, and a non-woven textile layer.

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TABLE VII. Waterproof insert requirements.

Characteristic	Requirement
Weight, oz./sq. yd., (maximum)	3.3
Thickness, inches, (maximum)	0.022
Moisture Vapor Transmission Rate, grams/square meter/24 hours, (minimum) Method B (hand-side to water) Method BW (hand-side to water)	 600 5000
Cut Strip Break Strength, pounds, (minimum) Warp Fill	 10 10
Seam Strength, pounds, (minimum)	6
Water Permeability after Cold Gelbo Flex, 5,000 cycles (0°F)	No leakage
Water Permeability after Wet Flex, 24 hours	No leakage
Water Permeability after Universal Wear Abrasion Test, 150 cycles	No leakage
Water Permeability after DEET (diethyltoluamide) insect repellent reagent	No leakage
Mullen Hydrostatic Resistance, psi, (minimum) Initial After DEET (diethyltoluamide) insect repellent reagent	 20 16
Chemical Penetration Resistance	No leakage

3.7 Components.

3.7.1 Curved tension lock 3/4-inch. The 3/4-inch curved tension lock shall be ITW Nexus, Part number GT LL 3/4-inch. The color of the tension lock shall be Foliage Green 504, Coyote 498 or as specified in the contract. Testing shall be as specified in 4.4.1 (see 6.8).

3.7.2 Tape wrist strap. The tape for the wrist strap shall be 3/4-inch wide conforming to Class 2 of MIL-PRF-17337. The long wrist strap shall be folded 1 3/4 (± 1/4) inch on one end and stitched at both ends of the fold. The unfolded raw end shall be stitched into the glove. The short wrist strap shall be folded in half and the raw edges stitched into the glove. The tape shall be cut in sufficient lengths to meet the specified finished lengths in Table VIII (see Figures 2 and 3). All tape raw edges shall be seared. The color shall be Foliage Green 504, Coyote 498 or as specified in the contract.

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TABLE VIII. Wrist strap lengths.

Sizes	Long wrist strap finished length (inches)	Short wrist strap finished length (inches)
Small	6 -1/2	1 -1/2
Medium	6 -1/2	1 -1/2
Medium - Wide	7	1 -1/2
Large	6 -7/8	1 -1/2
Large - Wide	7 -3/8	1 -1/2
X-Large	7	1 -1/2
X-Large – Wide	7 -3/8	1 -1/2
Tolerance	± 1/4	± 1/8

3.7.3 Glove hook. The glove hook shall be ITW Nexus, Part number 106-0100. The color shall be Foliage Green, 504, Coyote 498 or as specified in the contract. Testing shall be as specified in 4.4.1. The glove hook shall be placed on the side of the 1st digit on the right hand, and it shall be placed on the side of the 5th digit on the left hand. They shall be placed 1/2 – 3/4-inch from top of gauntlet (see 6.9).

3.7.4 Hook attachment material. A 3/8-inch wide ribbon or leather shall be used to attach the glove hook. The cut length shall be 2 (± 1/4)-inch, with 1/4-inch seam allowance. The distance of glove seam to glove hook on end item shall be 1/2 (± 1/4)-inch.

3.7.5 Elastic. The elastic webbing shall conform to MIL-W-5664, Type 1, Class 1, 3/4-inches in width. Elastic dimensions shall be as follows:

Size	Elastic Cut measurement (inches)	Finished Elastic Glove measurement (inches)
Small	3.5	3.75
Medium	3.75	4.0
Medium Wide	4.125	4.375
Large	3.75	4.0
Large Wide	4.375	4.625
X-large	4.125	4.375
X-large Wide	4.5	4.75

The elastic take-up for on the wrist side shall be 3/4-inch and placed across the entire wrist piece of leather, 1/2-inch from base of leather. The elastic shall provide proper cinch for snug fit (not too tight) at wrist and allows for easy don/doff (see 3.11.1.3) for stitch type.

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3.7.6 Thread. The thread for stitching and hemming shall be nylon conforming to A-A-59826, Type II, Class B, Size B, in color Foliage Green 504, Coyote 498 or as specified in the contract. The thread for stitching the lining unit and for attaching the lining to the waterproof insert shall be nylon conforming to A-A-59826, Type I or II, Class B, Size B, natural in color. As an alternate, cotton-covered polyester thread conforming to of A-A-50199, Tex Number 36-45 may be used for stitching the lining.

3.8 Finished glove.

3.8.1 Liner retention. The finished glove shall have no detachment/failure of the liner or waterproof insert when tested for retention of the glove liner as specified in 4.4.1.

3.8.2 Whole glove leak test requirement. The finished glove shall pass the whole glove leak test when tested as specified in 4.4.1.

3.9 Label. Each glove shall have a combination size and identification label, sewn-in at the wrist cuff on the inside of the glove, conforming to Type VI, Class 4 of MIL-DTL-32075. The label shall show colorfastness to laundering as specified therein. The label shall include the following information:

Gloves, Men's and Women's Intermediate Cold/Wet
Contract No.:
Contractor's Name:
Size:

3.9.1 Instruction slip. A printed slip of thin paper containing instructions as indicated below shall be inserted in each right hand glove.

Instructions

1. Gloves are not designed for heavy work such as rappelling or barbed wire handling, for hot work such as welding or changing machine gun barrels, or for use by POL handlers. Use other handwear for such activities.
2. Wear gloves alone or with Glove Inserts, Cold Weather, Lightweight. Do not wear with thicker inserts.
3. Fit gloves snugly but not tightly over the inserts.
4. Adjust strap for snug but not tight fit.
5. Gloves are available in 7 sizes: S – XL-WIDE.
6. Gloves have waterproof "breathable" linings but may become wet from perspiration - change glove inserts when damp, and dry wet gloves and glove inserts away from flames or extreme heat.
7. Clean shells by placing gloves on hands and washing outside with cold water and mild low-suds soap, rinsing thoroughly. Do not machine wash. Do not use detergents, solvents or bleach.
8. Treat outside of shells with fluorocarbon spray, do not use oils, wax, polish, or silicone.

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3.9.2 Bar code label/tag. Each item shall be individually bar coded conforming to Type VIII, Class 17 of MIL-DTL-32075. There shall be a 12 digit UPC number assigned for all NSN's by the contracting activity. The initials "UPC" shall appear beneath the code. The bar code for NSN and UPC type shall be a medium to high density and shall be located so that it is completely visible on the glove when it is folded and packaged as specified and so it causes no damage to the glove. This UPC code shall also be placed on all shipping cartons on which the NSN appears.

3.10 Patterns. Standard patterns (size medium) which show size, directional lines, placement marks, and notches for assembly, except where otherwise specified, will be furnished by the Government. The Government pattern shall not be altered in any way and shall be used to create the contractor's working patterns. Minor pattern modifications are permitted to accommodate manufacturing procedures; however, the design and finished measurements must be maintained. Pattern guidelines shall not be modified. Adhesive patterns are prohibited. The Government shall provide approval for any pattern changes that impact the appearance, serviceability, or dimensional requirements prior to incorporation into production.

3.10.1 List of pattern parts. Pattern parts shall be as listed in Table IX listed below.

TABLE IX. List of pattern parts.

Material	Pattern abbreviation	Nomenclature	Cut
Insulation	INSPB	Insulation Palm Back	2 (mirror)
	INSD3D4	Insulation D3 D4	2 (mirror)
	INSTPB	Insulation Thumb Palm Back	2 (mirror)
	INSTD2	Insulation Thin D2	2 (mirror)
Leather	D1PALM	D1 Palm	2 (mirror)
	D2D3F	D2 D3 Fourchette	2 (mirror)
	D3D4F	D3 D4 Fourchette	2 (mirror)
	D4D5F	D4 D5 Fourchette	2 (mirror)
	D5F	D5 Fourchette	2 (mirror)
	PALM	Palm	2 (mirror)
	PALMP	Palm Patch	2 (mirror)
Fabric (Cut Fill)	GAUPALM	Gauntlet Palm	2
	GAUPBACK	Gauntlet Back	2
	D1BACK	D1 Back	2
	BACK	Back	2
Nosewipe material	NOSEWIPE	Nose Wipe	2 (mirror)
Fleece (Cut warp)	WRISTLET	Wristlet	2

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3.11 Construction.

3.11.1 Stitches, seams, and stitching. Stitches, seams, and stitching types shall conform to ASTM D6193. Seam allowances shall be maintained with seams sewn so that raw edges, twists, pleats, or puckers will not result. All seams shall start and finish evenly. Ends of a continuous line of stitching shall be overlapped not less than 1/2-inch.

3.11.1.1 Type 301 stitching. Ends of all external seams and stitching, when not caught in other seams or stitching, shall be backtacked not less than 1/4-inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. If not backstitched, then separate 1/4-inch bartacking can be used to prevent unraveling.

3.11.1.1.1 Repairs of type 301 stitch. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2- inches back of the end of the stitching. When making these repairs, the ends of the stitching are not required to be backstitched.

b. Thread breaks or two more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/4-inch back of the defective area, continue over the defective area, and continue a minimum of 1/2-inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the material and re-stitching in the required manner.

3.11.1.2 Type 401 stitching. As an alternate, lining package (interior) can be sewn using Type 401 stitch. Ends of all external seams and stitching, when not caught in other seams or stitching, shall be backtacked not less than 1/4-inch. Thread tension shall be maintained so that there will be no loose stitching. If not backstitched, then separate 1/4-inch bartacking can be used to prevent unraveling. All repairs shall be made in accordance with 3.11.1.1.1 a. and b. Repairs to Type 401 stitching may be accomplished by use of stitch Type 301.

3.11.1.3 Zig-zag stitching. Two (2) rows of 304 zigzag stitch, seam type SSau-2, with 3/4-inch elastic on inside of glove assembly shall be used to secure the elastic take-up on wrist.

3.12 Finished measurements. The finished measurements for the glove shall be as specified in Table X.

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TABLE X. Finished measurement of gloves (inches).

Size	Overall length 1/	Width 2/
Small	10 -1/4	4 -1/2
Medium	10 -3/8	4 -3/4
Medium Wide	10 -3/4	5 -1/8
Large	11 -1/8	5 -1/8
Large Wide	11 -1/2	5 -3/8
X-Large	12	5 -1/8
X-Large Wide	12 -3/8	5 -1/2
Tolerance 3/	± 3/8	± 3/8

1/ The glove length shall be measured from tip of middle finger to edge of stretch fabric cuff, at the seam attaching the snow cuff. Measure on the palm side of the glove. Do not include the snow cuff in the length measurement. The glove shall be gently stretched lengthwise and then allowed to return to its natural state before measuring.
2/ The glove width shall be gently stretched out and measured across the width on the hand/palm portion, 1/2-inch down from the base of the little finger.
3/ The tolerance provided allows for the three dimensional pattern construction and the thickness of the insulation.

3.13 Toxicity. The finished textile materials shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.5.6. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.14 Workmanship. The finished glove shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in 4.4, shall not exceed the criteria specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color, finished measurements listed in Table X, finished defects listed in 4.4 and tested for the characteristics as specified in 4.4.1.

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4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.4.1 through 4.5 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether the components are as specified or operations and assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance.

4.3.2 Inspection conditions. Unless otherwise specified in this specification or applicable procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 End item visual examination. The finished gloves shall be examined for the defects listed in Table XI. The lot size shall be expressed in units of gloves. The sample unit shall be one pair of gloves and selection shall be by pairs. Defects for pairing shall be classified as a single defect.

TABLE XI. End item visual defects.

Examine	Defect	Classification	
		Major	Minor
Glove: Pairing	- Not matched and paired, right and left not same size, wide variation in appearance - Not tacked or joined together as specified	101	201
Shell or lining design	- Not as specified (incorrect material, pattern, etc.) - Poorly shaped - Bartacks missing, insecure, misplaced, not specified size, stitches loose or broken, bartack not serving intended purpose	102 103	202
Cleanness	- Spots or stains clearly noticeable affecting appearance		203
Leather: Color	- Not specified color - Color not uniform on grain side - Color does not completely penetrate leather from grain side through flesh side	104 105	204
Finish	- Not full grain Flesh side not smooth or contains areas of coarse loose fiber - Grain surface has application of finish - Embossing other than a fine hair cell pattern - Printed design on leather - Does not match standard sample for fine hair cell pattern	106 107 108 109	205

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TABLE XI. End item visual defects. – Continued

Examine	Defect	Classification	
		Major	Minor
Quality of leather	- Not clean; stain or foreign matter 1/	110	
	- Hard, boney, loose, spongy leather, hard scar, cut, hole (except a pinhole or needle hole), brittle, thin spot, brand, scratch, deep fat wrinkle, or grain damage which causes more than a 12 percent in total area of a side or more than a 6 percent loss in total area of a skin 1/	111	
Fabric	- Not specified color, streaky, shaded from side to side	112	
	- Any run, dropped stitch, snag, pull or slubby yarn	113	
	- Broken end	114	
	- Lacking elasticity, too tight or loose	115	
Membrane insert	- Delaminating	116	
Lining	- Facing delaminated		206
Tape	- Not type or color specified	117	
	- Hole, cut, tear or rip, float, smash or other defects affecting function		207
	- No affecting function	118	
Fleece Snow cuff	- Hole, cut, tear, smash, burn, drill hole, run, thin place, dye streak, color not as specified, mis-weave affecting appearance or serviceability	119	
Glove hook	- Not properly oriented	120	
	- Not functional	121	
	- Having any cracks, splits, or slices	122	
Construction, and workmanship; general (applicable to all components unless otherwise indicated herein)	- Component misplaced, operation omitted, or operation not properly performed	123	
	- Lining reversed so that tricot backing is oriented toward hand	124	
	- Component missing or not securely affixed	125	
	- Mend in leather or fabric (i.e., patch- not applicable to re-stitched seam repair)	126	
	- Needle chews likely to develop into a hole	127	

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TABLE XI. End item visual defects. - Continued

Examine	Defect	Classification	
		Major	Minor
Seams and stitching	Open seam 2/		
	- in single stitched seam	128	
	- in any row of a double stitched seam	129	
	- not repaired as specified (when applicable)		208
	- Not specified seam type	130	
	- Not specified stitch type	131	
	- Row of stitching omitted	132	
	- Loose stitch tension resulting in a loosely secured seam	133	
	- Tight stitch tension resulting in cutting of leather or breaking of stitches when normal pull is applied	134	
	- Ends of stitching not secured as specified		209
	- Gage of stitching irregular or not as specified, affecting appearance	135	
	- Part caught in unrelated row of stitching	136	
	- One or two stitches per inch less than specified		210
	- Three or more stitches per inch less than specified	137	
	- Not backstitched where required	138	
	More than the specified maximum number of stitches per inch:		
	- damaging materials	139	
	- not damaging materials		211

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TABLE XI. End item visual defects. - Continued

Examine	Defect	Classification	
		Major	Minor
Assembly detail	Glove not neatly laid off:		212
	- wrinkles in shell	140	
	- fingers not completely opened	141	
	- Inseam construction not where specified	142	
	- Seam at fingertip, special attention to detail	143	
	- Fourchettes missing from any finger or thumb	144	
	- Shell more than 1/4-inch longer than lining in any finger when on hand form	145	
	- Poorly assembled seriously affecting serviceability (finger distorted or twisted affecting comfort of wearer or seam of finger may come in contact with object, buckle backwards, etc.) 3/	146	
	- Waterproof insert and/or lining not securely fastened to shell fingertips	147	
	If adhesive is used, improper application of adhesive in fingertips:		
	- affecting dexterity or comfort (lumps, lining twisted, wrinkled or bunched, etc.)		213
	- Affecting appearance (stains on shell, etc.)	148	
	- Lining and/or insert not caught in cuff hem		214
	- Thread ends not trimmed throughout glove		215
	- Not constructed with specific number of pieces	149	
	- Difference in overall length between front and back of glove is more than 1/2-inch	150	
Label and Instruction Slip	- Omitted, incorrect, illegible, or misplaced; size and identification label not securely sewn in wrist hem		216
Label/tag	- Causes damage to the glove	151	
	- Label missing, incorrect, or illegible		217
	- Bar code omitted or not readable by scanner;		218
	- Human-readable interpretation (HRI) omitted or illegible;		219
	- Bar code not visible on folded, packaged item; bar code causes damage to the item		220

1/ Light well healed scratches or grub holes, light fat wrinkles, or slight stains shall not be classified as defects.
2/ A seam shall be classified as open when one (1) or more stitches joining a seam are broken or when two (2) or more consecutive skipped or run-off stitches occur.
3/ Any construction defect that results in a malformed glove such as a twisted finger or thumb resulting in any seam lying in the palmar area of the finger or thumb is a minor defect. When a finger or thumb is twisted and the seam is rotated to less than 45 degrees from the vertical and does not lie in the palmar area, it is a minor defect.

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4.4.1 Component/end item testing. The materials shall be tested for the characteristics listed in Table XII and the finished gloves shall be tested in accordance with Table XIII. The methods of testing as specified wherever applicable and as listed in Tables XII and XIII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished materials for all physical and chemical tests. The sample unit for glove testing shall be one (1) pair of finished gloves and selection shall be by pairs. The lot shall be unacceptable if one (1) or more sample units or the lot average fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (units)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE XII. Material component testing.

Material	Characteristic	Requirement Paragraph	Test Method
Goatskin	Thickness	3.6.1.1	ASTM D1814 1/
	Color	3.6.1.2	2/
	Colorfastness to crocking	3.6.1.3	AATCC TM8 3/
	Stitch tear strength	3.6.1.6	ASTM D4705
	Elongation	3.6.1.7	ASTM D2209
	Shrinkage Temperature	3.6.1.8	ASTM D6076
	Stiffness	3.6.1.9	ASTM D2821 and 4.5.1
	Water absorption resistance	3.6.1.10	4.5.2
	MVTR	3.6.1.11	ASTM D5052
	Hexane soluble material	3.6.1.12	ASTM D3495
	Chromic oxide	3.6.1.12	ASTM D2807
	pH value	3.6.1.12	ASTM D2810
	Total ash	3.6.1.12	ASTM D2617
Fabric, back of hand	Fiber Identification	3.6.2.1	AATCC TM20A (see 6.14)
	Color	3.6.2.1	2/
	Weight	3.6.2.1	ASTM D3776/D3776M
	Stretch	3.6.2.1	ASTM D4964 4/
	Water repellency	3.6.2.1	AATCC TM22
	Colorfastness to crocking	3.6.2.1	AATCC TM8 3/
	Tear Strength	3.6.2.1	ASTM D1424
Nosewipe	Fiber Identification	3.6.3	AATCC TM20A (see 6.14)
	Color	3.6.3	2/
	Weight	3.6.3	ASTM D3776/ D3776M
	Crocking	3.6.3	AATCC TM8 3/
	Tear Strength	3.6.3	ASTM D2261

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TABLE XII. Material component testing. - Continued

Material	Characteristic	Requirement Paragraph	Test Method
Fleece snow cuff	Fiber Identification	3.6.4	AATCC TM20A (see 6.14)
	Color	3.6.4	2/
	Weight	3.6.4	ASTM D3776/ D3776M
	Colorfastness Crocking	3.6.4	AATCC TM8 3/
	Stretch	3.6.4	ASTM D4964
	Dimensional Stability	3.6.4	AATCC TM135
	Water repellency	3.6.4	AATCC TM22
Backing & Facing lining	Fiber Identification	3.6.5.1	AATCC TM20 (see 6.14)
	Color	Table V	2/
	Weight	Table V	ASTM D3776/D3776M
	Bursting strength	Table V	ASTM D3787
Index finger lining	Fiber Identification	3.6.5.1	AATCC TM20A (see 6.14)
	Color	Table V	2/
	Weight	Table V	ASTM D3776/D3776M
	Bursting strength	Table V	ASTM D3787
Finished lining	Thickness	3.6.5.3	5/
	Drape Stiffness	3.6.5.3	ASTM D1388 Opt A
	Water vapor transmission	3.6.5.3	ASTM E96/E96M
Waterproof Insert (laminate)	Weight	Table VII	ASTM D3776 D3776M 6/
	Thickness	Table VII	ASTM D1777 7/
	MVTR:		
	Method B	Table VII	ASTM E96/E96M 8/
	Method BW	Table VII	ASTM E96/E96M 9/
	Cut Strip Break strength	Table VII	ASTM D5035
	Seam Strength	Table VII	ASTM D1876 10/
	Water Permeability after Cold Gelbo Flex	Table VII	ASTM F392/F392M 11/ ASTM D751 12/
	Water Permeability after Wet Flex	Table VII	ASTM D751 13/, 12/
	Hydrostatic Resistance		
	Initial	Table VII	ASTM D751 14/
	After DEET	Table VII	4.5.3 and ASTM D751 14/
	Chemical Penetration Resistance	Table VII	ASTM F903 15/
	Water Permeability after Universal Wear Abrasion Test	Table VII	ASTM D3886 16/ ASTM D751 12/
	Water Permeability after DEET	Table VII	4.5.3 and ASTM D751 12/

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TABLE XII. Material component testing. - Continued

Material	Characteristic	Requirement Paragraph	Test Method
Tension lock	Color	3.7.1	2/
Tape (wrist strap)	Color	3.7.2	2/
Glove hook	Color	3.7.3	2/
Thread	Color	3.7.6	2/

1/ (Note: 1 ounce = 1/64-inch).
2/ The color and appearance of the materials shall match the standard sample when viewed using the AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, Option C (see 6.15), with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.
3/ Rated using the AATCC EP8, Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale.
4/ Test Method ASTM D4964 shall be run at 20-inches per minute and the specimens cycled 5 times.
5/ The finished lining thickness shall be measured on panels using a 12-inches x 12-inches calibrated Measure-Matic Thickness Gauge, or equivalent, under a pressure of 0.002 pounds per inch (psi).
6/ Five (5) 3 -1/2 inch diameter circle specimens shall be tested. The individual and mean weight of the five specimens shall be reported to the nearest 0.1 ounces per square yard.
7/ Five (5) 3 -1/2 inch diameter circle specimens shall be tested. Pressure foot shall be 1.129-inches diameter applying a load of 0.125 lbs./in^2. The individual and mean thickness of the five (5) specimens shall be reported to the nearest 0.001 inches.
8/ Face side for fabric, back of hand and the hand side of the insert shall face the water. [Please note the insert may be inverted in use; please refer to end item.] The free stream air velocity shall be 550 (± 50) fpm as measured two inches above the fabric specimen. The airflow shall be measured at least 2-inches from any other surface. The test shall be run for 24 hours and weight measurements shall be taken at only the start and completion of the test. At the start of the 24-hour test period, the air gap between the water surface and the back of the specimen shall be 3/4 (± 1/16)-inch. Five (5) specimens shall be tested.
9/ Face side for fabric, back of hand and the hand side of the insert shall face the water. [Please note the insert may be inverted in use; please refer to end item.] The free stream air velocity shall be 550 (± 50) fpm as measured 2-inches above the fabric specimen. The airflow shall be measured at least 2 inches from any other surface. The test shall run for 2 hours and weight measurements shall be taken at only the start and completion of the test. Five (5) specimens shall be tested. The specimen shall be sealed in any manner which prevents wicking and/or leaking of water out of the cup.
10/ A minimum of three (3) specimens shall be tested. Reference ASTM D1876 with the following modifications: Specimens (1-inch x 6-inch) shall be cut from palm of insert to include sealed seam in center of specimen, no conditioning required, tension testing machine head speed 10-inches per minute, report maximum peeling load only.
11/ One (1) specimen, 8-inches by 12-inches, shall be cut from the sample unit with the 8- inch side in warp direction and one specimen with the 8-inch side in the fill direction. The specimen shall be conditioned at 0°F for one (1) hour and flexed as specified in

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ASTM F392/F392M except the specimen shall not be aged, the short edges shall not be heat sealed or otherwise joined, and the specimen shall be flexed for 5,000 cycles. At the end of the flexing, the specimen shall be removed from the test chamber and conditioned prior to testing for water permeability. Test two sites for leakage at completion of flex using same conditions and inspection criteria as
Footnote 9.
12/ The water permeability shall be tested in accordance with ASTM D751, Hydrostatic Resistance, Procedure B, with a fixed hydrostatic head of 50 cm minimum shall be held for 3 minutes minimum, testing (come into contact the water) the side of the sample (insert) facing the shell. [Please note the insert may be inverted in use; please refer to end item.] Three (3) specimens minimum shall be tested. The report shall include only measurement of the appearance of the water droplets. Leakage is defined as the appearance of one or more droplets anywhere within the 3-1/2-inch minimum diameter test area. In order to use Suter Hydrostat tester after abrasion, one will have to fabricate a specimen holder with 3-1/2-inch minimum diameter.
13/ Test one (1) sample, 14-inches x full width (28-inches minimum). Specimen shall be agitated using the "normal" cycle in an automatic home laundering machine except that the machine shall be capable of continuous agitation [AATCC TM135, without soap. The water level shall be maintained at 16 (± 0.5) gallons and water temperature shall be 32 (±-9) °C. Additional fabric shall be added to create a load of 2 (± 0.2) pounds.] Upon completion, the specimen shall be air dried prior to testing for leakage in accordance with Footnote 12.
14/ Test five (5) samples. ASTM D751 (Hydrostatic Resistance Procedure A1). No restraining or supporting fabrics are to be used during the testing. Test the side of the sample (insert) facing the shell. [Please note the insert may be inverted in use; please refer to end item.]
15/ Test five (5) samples using exposure Procedure C. Expose shell side of laminate. Surrogate gasoline Fuel C as defined by ASTM D471.
16/ Test five (5) samples abrading the side of insert facing shell. The abrasion test shall be conducted in the multidirectional mode using 0 grit emery paper as the abradant. Use a solid rubber diaphragm without electrical contact (grounding) pin. The diaphragm shall have a smooth surface without bumps. Use diaphragm inflated to 4 (± 0.25) psi. A load of 1 lb. shall be applied to the abradant. The test mode shall be 100 double strokes per revolution. Do not change the abradant paper. Once abrasion is complete, specimens shall be tested using water permeability test method as described in Footnote 12.

TABLE XIII. End item (glove) testing.

Material	Characteristic	Requirement Paragraph	Test Method
Finished Glove	Liner retention test	3.8.1	4.5.4
	Whole glove leak test	3.8.2	4.5.5

4.5 Methods of inspection.

4.5.1 Stiffness test.

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a. Degree of Stiffness before soaking. Tested according to ASTM D2821 except the specimens shall be pre-flexed as follows:

1. Loosen knob (G). Using knob (H) slowly rotate the top dial (K) in a counterclockwise direction until the specimen clamp indicator (I) shows that the specimen has been twisted 90°. Use a uniform rate of rotation that will take 4 to 5 seconds to twist the specimen through 90°.

2. Reverse rotation to the clockwise direction and continue rotating slowly until the specimen clamp indicator (I) has passed the original reference line indicated on the lower dial (J) by approximately 45°. Again reverse the direction of rotation and slowly realign the specimen clamp indicator (I) with the original reference line indicated on the lower dial (J).

3. Lock top dial (K) by turning knob (G) until tight, then reset dial (K) to read 270°.

4. Repeat the flexing operation by following steps 1, 2, and 3. When the second flexing cycle is completed, loosen knob (G), turn upper dial (K) slowly (4 to 5 seconds for 90° rotation) in a counterclockwise direction, using knob (H), until the specimen clamp indicator (I) shows 90° twist. Read the angle of torsion from the upper dial (K) and record the value to the nearest 5 degrees. The angle of torsion of each specimen shall be reported to the nearest 5 degrees.

b. Degree of Stiffness after soaking. If the stiffness requirements specified for before soaking are not met, then do not test for stiffness after soaking. Any individual specimen exceeding the maximum stiffness values specified for before soaking shall not be tested for stiffness after soaking. The specimens shall be immersed in 1,000 mL of distilled water at 140 (± 5) °F for 20 (± 2) hours. Complete immersion of the specimens shall be accomplished by placing a floating cover on the vessel used for the immersion of the specimens. After the required immersion period, the specimens shall be removed from the water and reconditioned in accordance with standard atmospheric conditions as specified in ASTM D2821. The reconditioned specimens shall be placed flesh side down on a rule graduated in decimal inches and the distance between the bench marks shall be measured to the nearest 0.05-inch. If the distance is 3.4 to 3.6-inches, test the specimen in accordance with ASTM D2821 and the exceptions. If the distance is 3.2-inches or less, the specimen shall be reported as a test failure. If the distance between benchmarks is 3.25 to 3.35-inches, the specimen shall be stretched before testing. To stretch the specimen, place the specimen on the rule and align one of the benchmarks with a division line on the rule. Hold the specimen against the rule by placing the fingertip of one hand on the aligned benchmark. With the thumb and index finger of the other hand, lightly grip the specimen near the aligned bench mark, then slide the finger and thumb along the specimen to the other bench mark while exerting pressure on the specimen that is sufficient to obtain the required distance between the bench marks. If the required distance between the benchmarks has been obtained (3.4 inches to 3.6 inches), test the specimen in accordance with ASTM D2821 and the exceptions (see Footnote 3). If the distance between the benchmarks is greater than 3.6-inches discard the specimen and test a new specimen selected from the same 12 x 12-inch test sample of leather from which the discarded

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specimen was obtained. The new specimen shall be tested for stiffness before and after soaking. The test results obtained when the discarded specimen was tested for stiffness before soaking shall not be reported.

4.5.2 Water absorption resistance test. One 3-inch diameter specimen shall be tested from each 12 x 12-inch test area. Prior to testing, the specimen shall be allowed to reach moisture equilibrium in the standard conditions environment. A stainless steel 500 mL, 3-inch diameter, 5-inch long container with a rubber gasket and cover shall be used to test the specimen. The specimen shall first be weighed (to the nearest 0.01 g) and then shall be placed inside the cover with the flesh side of the specimen next to the inside of the cover. Place the rubber gasket inside the cover and against the grain surface of the leather specimen. Add 100 mL of distilled or deionized water at 23 °C to the container. Secure the cover with the specimen and gasket to the container. Invert the container so the water is against the grain surface of the specimen. Place the inverted container on a flat surface and allow it to remain in the inverted position for 30 minutes. At the end of 30 minutes place the container upright, remove the cover, and place it on a flat surface. The inside of the cover shall face the flat surface. Tilt the cover so the outer edge of the bottom is raised 1-1.5 inches from the flat surface, and allow it to drain for 5 minutes. After 5 minutes immediately remove the specimen from the cover, hold the specimen by its edge and shake briskly once to remove any free moisture from the surface of the specimen. Immediately re-weigh the specimen to the nearest 0.01 g, and calculate the percent water absorption as follows:

$$\text{Percent water absorption} = \frac{(W\,2 - W\,1) \times 100}{W\,1}$$

W1 = Original weight of conditioned specimen
W2 = Weight of specimen after being subject to water

4.5.3 Water absorption and hydrostatic resistance after DEET. The specimen, 4-inches X 4-inches minimum, shall be laid flat on a glass plate. Three (3) drops of the test liquid DEET shall be applied to the center of the specimen. Test the side of the sample (insert) facing the shell. A glass plate of the same dimensions shall be placed on the specimen and a four-pound weight placed in the center of the glass plate of the assembly. After 16 hours, remove the specimen and test immediately for hydrostatic resistance or water permeability, as required.

4.5.3.1 (DEET) insect repellent reagent. The insect repellent reagent should be a solution of 75% by weight (min.) of diethyltoluamide and the remainder denatured alcohol. The diethyltoluamide component of the solution shall be a technical grade and contain N, N-diethyl-m-toluamide of not less than 95% meta-isomer. The denatured alcohol component of the solution shall be ethanol, U.S.P. 94.9% by volume in accordance with Code of Federal Regulations 27 CFR 21, Formula 40 (see 2.2.2). The insect repellent shall be registered with the U.S. Environmental Protection Agency (EPA) in accordance with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) 40 CFR Part 162 (see 2.2.2).

4.5.4 Liner retention test. Test a minimum of three (3) finished gloves. Each digit of the glove shall be tested in accordance with 4.5.4.1 and 4.5.4.2.

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4.5.4.1 Liner retention test apparatus. Liner retention shall be evaluated with the use of locking forceps and a force-measuring gauge.

4.5.4.2 Test procedure. The locking forceps shall be attached to the inner liner of the digit to be tested ensuring that an unattached liner or the outer shell is not grabbed. The hook of the force gauge shall be looped around the locking bridge of the forceps. The digit of the glove shall be gripped ensuring the inner liner is not impeded. The force gauge shall be pulled until 6 pounds of force registers on the dial and then released. Each digit shall be inspected for indication of detachment of inner liner and the waterproof insert. Results shall be recorded and reported as pass or fail. Failure of any digit of any glove shall constitute failure. Glove shall be permitted to be cut open to verify detachment.

4.5.5 Whole glove leakage testing. The finished glove shall be tested for leakage in accordance with 4.5.5.1 and 4.5.5.2. The lot size shall be expressed in units of gloves. The sample unit shall be one (1) glove, and the selection shall be by pairs. Any test failure shall be cause for the rejection of the lot.

4.5.5.1 Testing apparatus. The test apparatus shall consist of a 3-inch inside diameter tubular rigid polyvinyl chloride test chamber approximately 12-inches in length. The chamber shall be capped approximately 2-inches from the bottom with the cap having 1/2-inch diameter plastic air escape tube approximately 3-inches long. The test chamber shall be oriented vertically and placed in a container filled with approximately 1-inch of water such that the escape tube vents to the outside of the chamber, with the end of the tube under the water. The top of tube shall be beveled at a 45° angle on the inside. The top plug shall consist of polyethylene, approximately 4 inches in height, and 4-inches in diameter at the top tapering to approximately 2-inches in diameter at the bottom. An air passage shall be provided vertical through the plug and connected via nominal 1/2-inch tubing through a gage and air regulator to an oil-free compressed air source. A mechanism shall be provided that is capable of reproducibly clamping the plug into the end of the tube such that a seal is provided. An equivalent to the apparatus as mentioned above shall be qualified by US Army Combat Capabilities Development Command (CCDC), Soldier Center (see 6.13).

4.5.5.2 Testing procedure. The glove shall be inserted into the test chamber to near the cuff. The plug shall be inserted into the cuff of the glove and clamped such that a seal that holds not less than 4 pounds per square inch is formed with the beveled edge of the test chamber. Air shall be provided to the chamber to a level of 4 psi for a period of not less than 5 seconds. Appearance of air bubbles at 5 seconds constitutes failure.

4.5.6 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the textile material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.13) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 6.2).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gloves are intended to provide environmental protection in intermediate cold and wet climates (0°F to 40°F temperature range). They may be worn alone or over lightweight glove inserts. The gloves are not intended for heavy work.

6.2 Acquisition requirements. Acquisition documents will specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2.1).
c. The specific issue of individual documents referenced (see 2.2).
d. When a first article is required (see 3.1, 4.2).
e. Conformance inspection acceptance quality limits (AQLs) (see 4.3)
f. When toxicity testing is required (see 3.13, 4.5.6)
g. Inspection conditions (see 4.3.1)
h. Packaging requirements (see 5.1).

6.3 Shade standard. Individual color chips and SAE-AMS-STD-595 are available from SAE International, http://www.sae.org (see 3.2).

6.5 Fabric, back of hand. Fabric found to meet the requirements for the back of the hand is available from Tweave, Inc., 38 Barrows Street, P.O. Box AV, Norton, MA 02766 (see 3.3.2).

6.6 Fabric, fleece snow cuff. Fabric, Fleece Snow Cuff is available from Polartec LLC, 310 Industrial Dr. SW, Cleveland, TN 37311.

6.7 Waterproof insert. The waterproof insert will be securely affixed inside the glove, such that it cannot be pulled out when hand is removed. Each finger will be inspected for indication of detachment of inner liner and the waterproof insert. To ensure non-detachment, the waterproof insert may be sewn to insulation package using the integral tabs that exist at the fingertips and alongside the thumb (digit one) and pinky (digit five). The insert is then inverted over the insulation package and is affixed inside the glove shell using pressure sensitive adhesive tabs,

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glue, etc. If pressure sensitive adhesive tabs are used, then it is recommended to place tabs at each fingertip and alongside the base of the trigger finger (digit two). This extra tab near the base of the trigger finger keeps the seam of the insert aligned with the side of the finger and out of the way of the finger pad area.

6.8 Curved tension lock 3/4 inch. Curved tension lock meeting the specified requirements is available from: ITW Nexus, 195 Algonquin Ave, Des Plaines, IL 60016. Military Sales (see 3.7.1).

6.9 Glove hook. The glove hook meeting the specified requirements is available from: ITW Nexus, 195 Algonquin Ave, Des Plaines, IL 60016. Military Sales (see 3.7.3).

6.10 Nosewipe. Nosewipe material meeting the specified requirements is available from: Spectro Coating Corp., 101 Scott Drive, Leominster, MA 01453 (see 3.6.3).

6.11 Thermal insulation. The US Navy Clothing and Textile Research Facility at the US Army Combat Capabilities Development Command (CCDC), Soldier Center, Natick, MA is capable of performing the required test (see 4.5.4).

6.12 Nujol. Nujol is a mineral oil available in most drug stores and has been found to be appropriate for use in testing the leather for oil absorption resistance. Nujol is the trademark of Plough, Inc.

6.13 Testing apparatus. Note that the Whole Glove Leak Tester is covered by US Patent 4,776,209 through July 13, 2007. Gore Enterprise Holdings is the owner by assignment of this patent. A non-royalty-bearing license (only) for gloves manufactured per MIL-DTL-44419 for the US Government may be obtained by contacting W.L. Gore & Associates, Inc. (Contact: Military Inside Sales/Glove Products (410) 392-3600.) Descriptions and drawings for constructing the tester are contained in the patent.

6.14 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.15 Visual shade matching. In 2019, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 ($\pm 5^{o}$) angle relative to the horizontal. The observer will view the specimen at a 90° angle, relative to the plane of the specimens.”

6.16 Subject term (key word) listing.

Enviromental
Handwear
Hand protection
Waterproof

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6.17 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

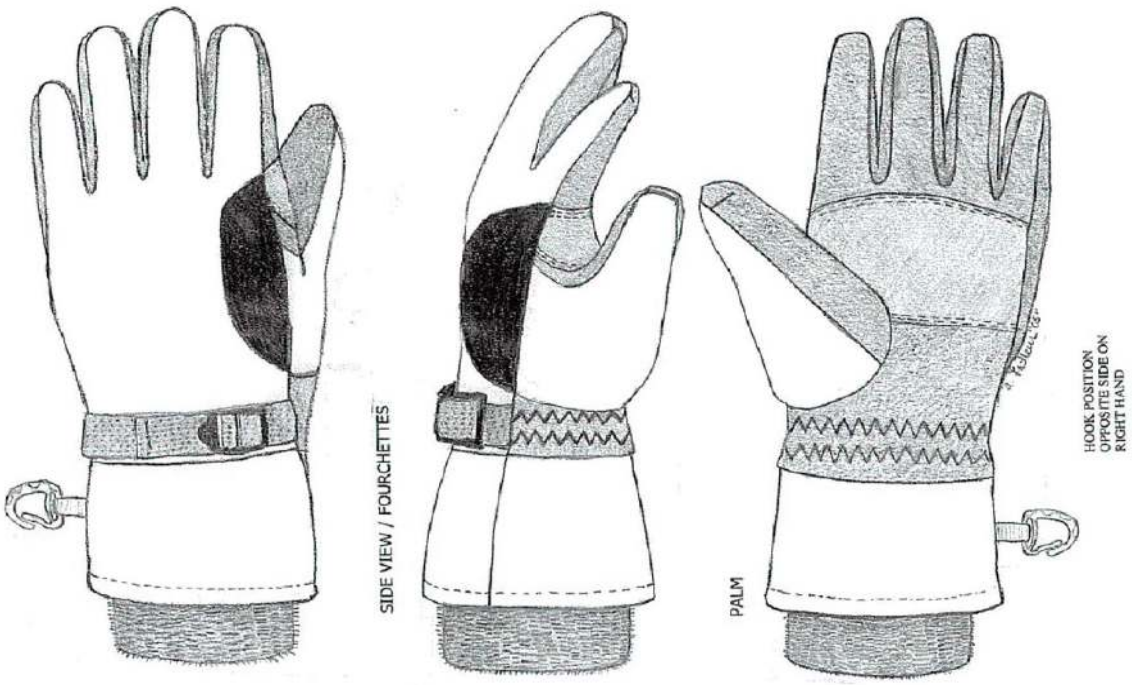
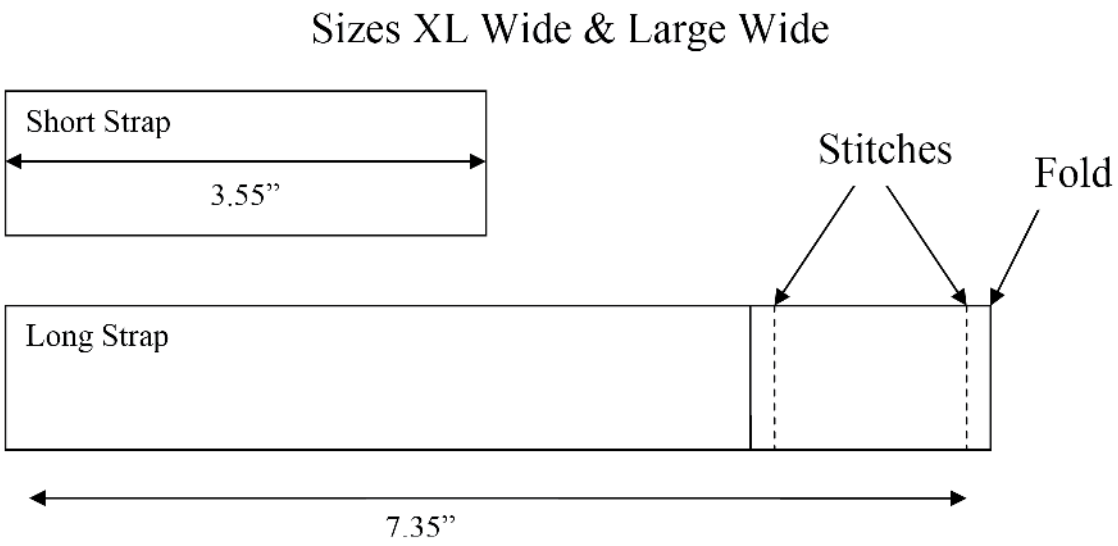


FIGURE 1. Intermediate cold/wet glove.

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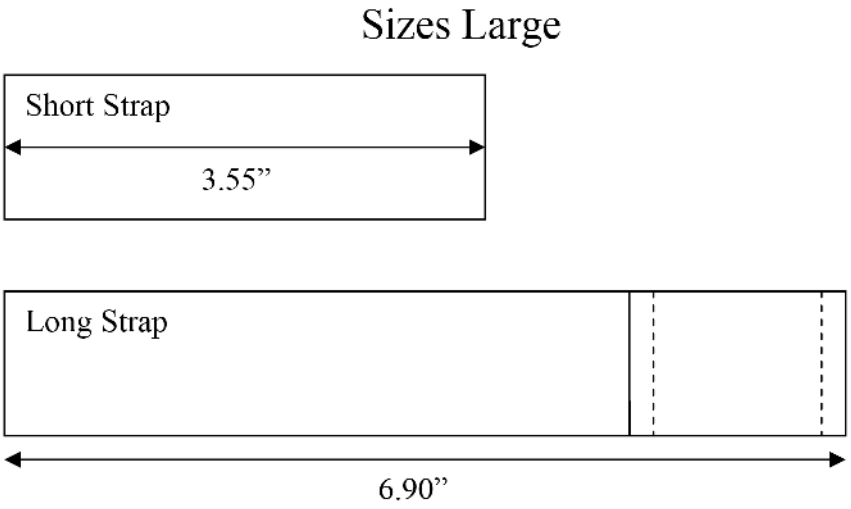
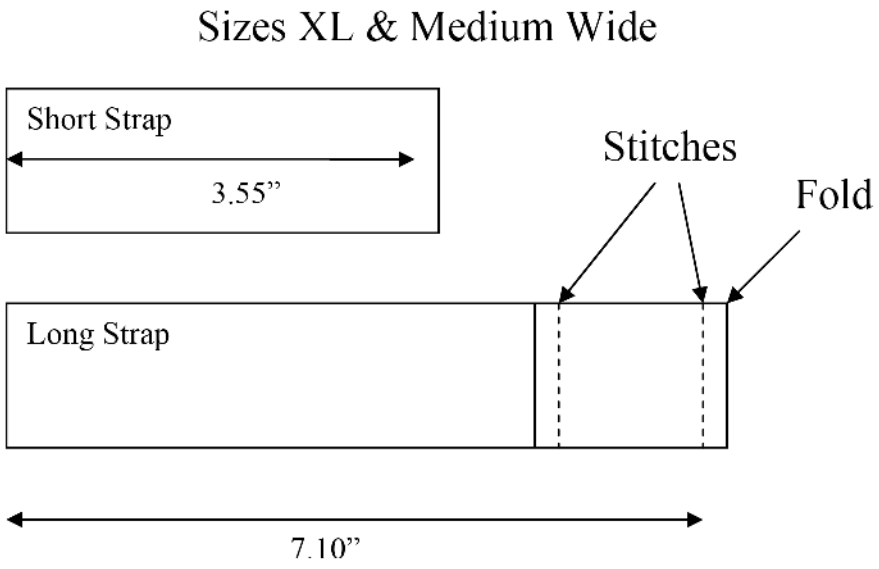


FIGURE 2. Tape, wrist strap (sizes L, L-wide and XL-wide).

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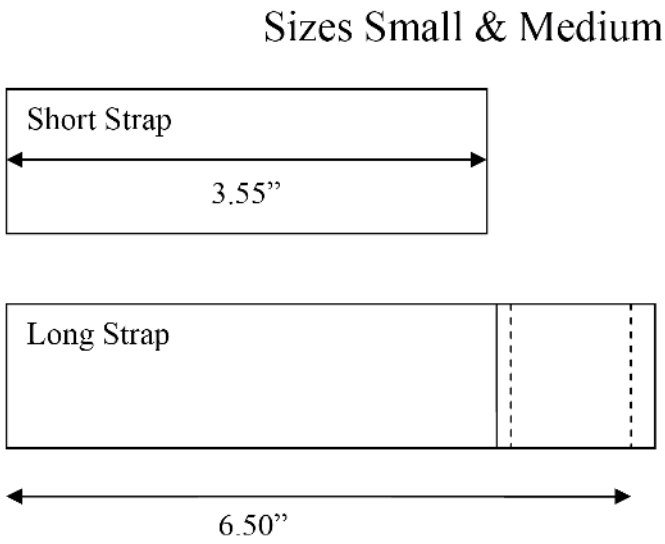


FIGURE 3. Tape wrist strap (sizes S, M, M-wide, and XL).

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Custodians:
Army – GL
Navy – NU
Air Force – 11

Review Activities:
Army – MD
Navy –CG1, MC

Preparing Activity:
DLA - CT

(Project: 8415-2020-052)

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